SHARK OF THE WIRE I

Challenge Description:

We intercepted a suspicious local transmission on a developer's machine. The communication was brief. Almost silent. Can you uncover what was transmitted?

Solution:

This is a standard packet capture file, just open it with WireShark, or TCPDUMP or even run strings on it with a grep for TACHYON:

```
mrsuave@mrsuave-HP-Laptop-15-bs0xx:~/Downloads$ strings challenge.pcap | grep TACHYON
TACHYON{n3tw0rking_is_4un}
```

Flag:

TACHYON{n3tw0rking_is_4un}

SHARK OF THE WIRE II

Challenge Description:

The sender wants to outsmart you, can you outsmart the sender? The intercepted transmission wasn't as straightforward this time. Instead of a single clean message, the data appears to have been broken into multiple fragments before being sent. Each piece may not make sense on its own.

Solution:

This is again a packet capture file, but the flag is encrypted with base64 and parts of the flag are sent in chunks and caught in the packet capture file.

Running strings won't help here:

```
mrsuave@mrsuave-HP-Laptop-15-bs0xx:~/Downloads$ strings chall.pcap
UVEFDSFlP
TntmcjRnb
TNudDNkXz
P]8P
FzX24wXzR
1.}S
KnK
1.}S
KnK
11bn0=
KnQ
1.}S
1.}S
1.}S
KnR
```

However, you can take out the parts one by one, concatenate them into:
VEFDSFlPTntmcjRnbTNudDNkXzFzX24wXzR1bn0=
this payload and then convert this into base64.

(Some trial and error might be involved in getting the right strings , for example 11bn0= is not directly part of string, only 1bn0= is, the first 1 is just part of TCP Capture dump.)

```
reading from file chall.pcap, link-type NULL (BSD loopback), snapshot length 524288
10:26:27.125578 IP6 ip6-localhost.59462 > ip6-localhost.4444: Flags [S], seq 3246872271, win 65535, options [mss 16324,nop,wscale 6,nop,nop,TS val 3631915864
ecr 0,sackOK,eol], length 0
`        ...,.@.................................F.\..V..........4....?........
.z.X........
10:26:27.125609 IP6 ip6-localhost.4444 > ip6-localhost.59462: Flags [R.], seq 0, ack 3246872272, win 0, length 0
`        ....@.................................\.F......V.P........
10:26:27.125664 IP localhost.59463 > localhost.4444: Flags [S], seq 1316192444, win 65535, options [mss 16344,nop,wscale 6,nop,nop,TS val 2031818062 ecr 0,sac
kOK,eol], length 0
E..@..@.@...........G.\Ns...........4....?........
y..N........
10:26:27.125699 IP localhost.4444 > localhost.59463: Flags [S.], seq 224115697, ack 1316192445, win 65535, options [mss 16344,nop,wscale 6,nop,nop,TS val 2477
74037 ecr 2031818062,sackOK,eol], length 0
E..@..@.@...........\.G.[..Ns.......4....?........
...Uy..N....
10:26:27.125709 IP localhost.59463 > localhost.4444: Flags [.], ack 1, win 6380, options [nop,nop,TS val 2031818062 ecr 247774037], length 0
E..4..@.@...........G.\Ns...[.......(.....
y..N...U
10:26:27.125719 IP localhost.4444 > localhost.59463: Flags [.], ack 1, win 6380, options [nop,nop,TS val 247774037 ecr 2031818062], length 0
E..4..@.@...........\.G.[..Ns.......(.....
...Uy..N
10:26:27.125735 IP localhost.59463 > localhost.4444: Flags [P.], seq 1:10, ack 1, win 6380, options [nop,nop,TS val 2031818062 ecr 247774037], length 9
E..=..@.@...........G.\Ns...[.......1.....
y..N...UVEFDSFlP

10:26:27.125743 IP localhost.59463 > localhost.4444: Flags [F.], seq 10, ack 1, win 6380, options [nop,nop,TS val 2031818062 ecr 247774037], length 0
E..4..@.@...........G.\Ns...[.......(.....
y..N...U
10:26:27.125750 IP localhost.4444 > localhost.59463: Flags [.], ack 10, win 6380, options [nop,nop,TS val 247774037 ecr 2031818062], length 0
E..4..@.@...........\.G.[..Ns.......(.....
...Uy..N
10:26:27.125755 IP localhost.4444 > localhost.59463: Flags [.], ack 11, win 6380, options [nop,nop,TS val 247774037 ecr 2031818062], length 0
E..4..@.@...........\.G.[..Ns.......(.....
```

Alternatively you could also use tcpdump or wireshark to check the packets that were pushed (have the P tag) and then retrieve text from them, to get the same payload:
VEFDSFlPTntmcjRnbTNudDNkXzFzX24wXzR1bn0=

Then decode using base64:

```
mrsuave@mrsuave-HP-Laptop-15-bs0xx:~/Downloads$ echo "VEFDSFlPTntmcjRnbTNudDNkXzFzX24wXzR1bn0=" | base64 -d
TACHYON{fr4gm3nt3d_1s_n0_4un}mrsuave@mrsuave-HP-Laptop-15-bs0xx:~/Downloads$
```

Flag: TACHYON{fr4gm3nt3d_1s_n0_4un}